

Descriptive statistics and plots

Overview

Plots and statistics of categorical data

Plots and statistics of quantitative data:

- Measures of central tendency

If there is time:

- Measures of spread
- Two quantitative variables

Announcement: Homework 2

Homework 2 is available

It is due on Gradescope by 11pm on Monday July 13th

- Question 4 involves reading a short article and commenting on it, so you can get started on this right away

How did homework 1 go?

Questions/comments about anything?

Review: Data frames

Data frames contain structured data

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
3	38	thin	anything	socially	NA	graduated from masters program
4	23	thin	vegetarian	socially	NA	working on college/university
5	29	athletic	NA	socially	never	graduated from college/university
6	29	average	mostly anything	socially	NA	graduated from college/university

Review: OK Cupid data



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BigDaddyC_taco
21 / M / Straight / Single
Chicago, Illinois

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[About](#)[Photos](#)[Questions](#)[Personality](#)

My self-summary

I'm a young, ambitious and outgoing individual. I love traveling, having recently been to South America and through the southern states on a road trip with friends. I'm a very caring/emotional person. I enjoy anything artistic and always up for new activities. Also, I've been told I'm too perfect.

What I'm doing with my life

- Working two marketing jobs in downtown and Lincoln Park areas of Chicago.
- Full-time student at DePaul University studying Marketing/Sales.
- Volunteer on South Side of Chicago (Pilsen, Little Village & Englewood).
- Writer for my blog, The Plaid Tie

My Details

Last Online	Online now!
Ethnicity	Hispanic / Latin
Height	6' 0" (1.83m).
Body Type	Fit
Diet	Mostly anything
Smokes	No
Drinks	Rarely
Drugs	Never

Review: Data frames

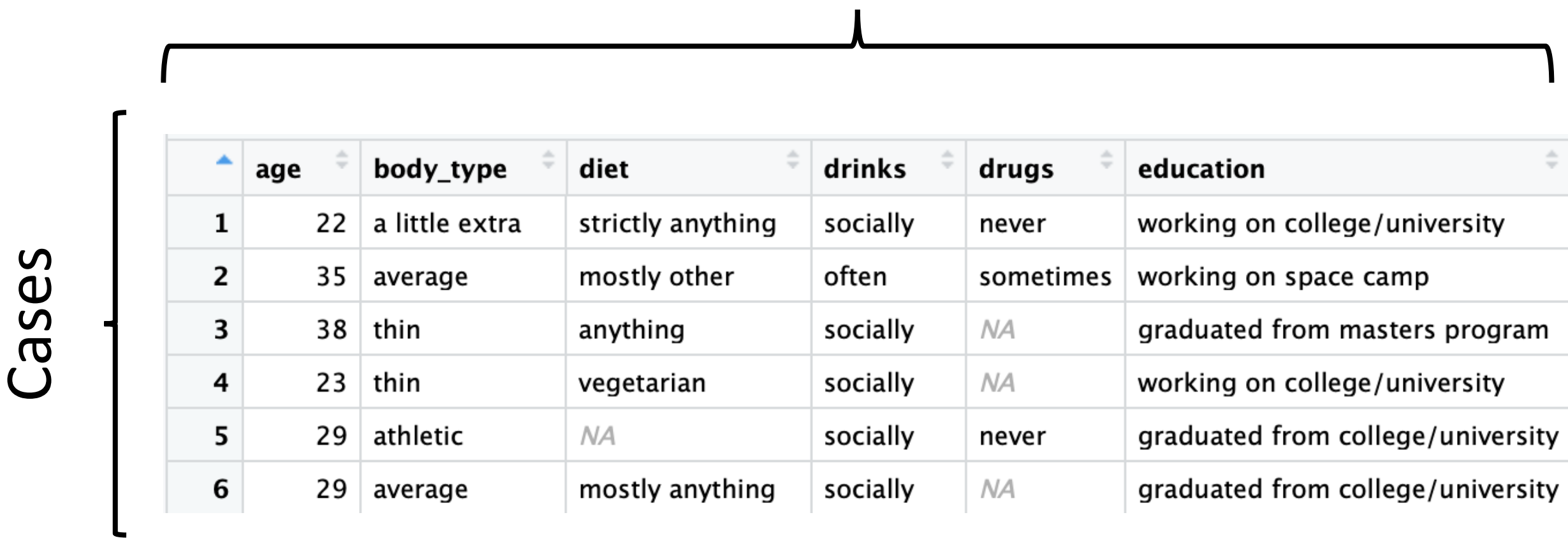
Data frames contain structured data

- > `library(SDS1110)`
- > `download_data("profiles_revised.csv")` # only needs to be run once
- > `profiles <- read.csv("profiles_revised.csv")`
- > `View(profiles)` # the `View()` function only works in R Studio!

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
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Review: Data frames

Variables



	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
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Cases

Review: Data frames

Quantitative Variable

Categorical Variable

Cases
(observational units)

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
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Review: Data frames

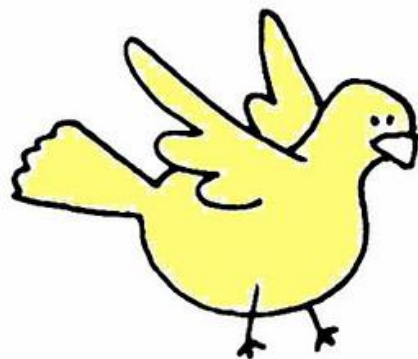
We can extract the columns of a data frame as vector objects using the \$ symbol

```
> the_ages <- profiles$age
```

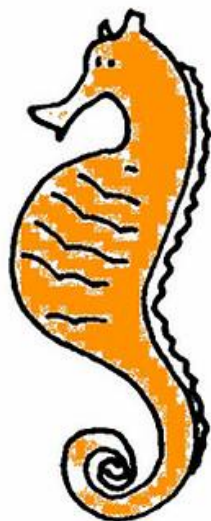
Can you get the average (mean) of the ages of users in this data set?

```
> mean(the_ages)
```

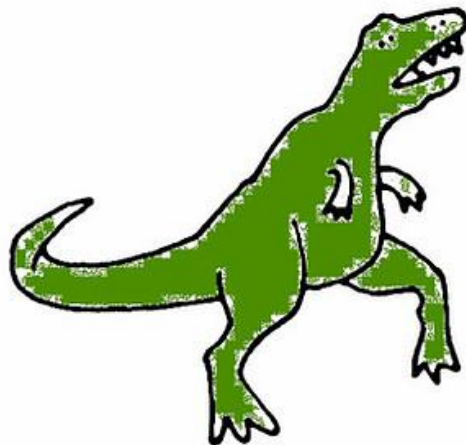
CATEGORICAL DATA:



I am a bird.
I am yellow.
I am awesome.



I am a seahorse.
I am orange.
I am super awesome.



I am a T-rex.
I am green.
I am extinct.

Categorical variables

A categorical variable assigns each observation to one of k groups

Which variables in the profiles data frame are categorical?

- Is age a categorical variable?

For categorical variables, we usually want to view:

- **Frequency table**: How many items are each category
- **Relative frequency table**: The proportion (or percentage) of items in each category

$$\text{Proportion in a category} = \frac{\text{number in that category}}{\text{total number}}$$

Frequency table

We can use the `table()` function to create a frequency table

```
# Get vector of drinking behavior
```

```
drinking_vec <- profiles$drinks
```



```
# Create the frequency table
```

```
drinks_table <- table(drinking_vec)
```

```
drinks_table
```

Category	Frequency
No data	2985
desperately	322
not at all	3267
often	5164
rarely	5957
socially	41780
very often	471
total (n)	59946

Relative frequency table

We can create a relative frequency table using the function:

```
prop.table(my_table)
```



Note: this argument is a table not a vector of data!

Can you create a relative frequency table for the drinking behavior of the people in the okcupid data set?

```
drinks_table <- table(profiles$drinks)
prop.table(drinks_table)
```

Category	Frequency
No data	0.05
desperately	0.005
not at all	0.054
often	0.086
rarely	0.099
socially	0.697
very often	0.008
total	1

Bar plots

(pun intended?)

We can plot the number of items in each category using a bar plot

```
barplot(my_table)
```

Can you create a bar plot for the drinking behavior of the people in the okcupid data set?

```
drinks_table <- table(profiles$drinks)
```

```
barplot(drinks_table)
```

Let's try it in RStudio!

Details matter!

Can you figure out how to label the axes?

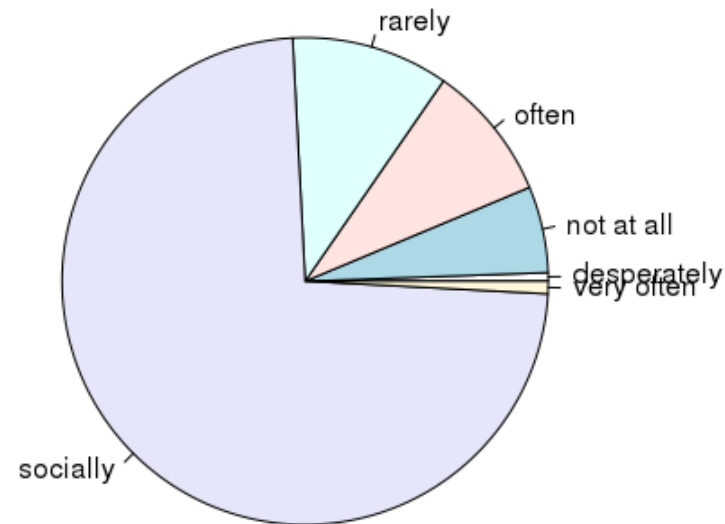
- A: ? barplot
- A: xlab and ylab!

```
barplot(drinks_table,  
        ylab = "Count",  
        xlab = "Type of drinker",  
        main = "Counts of different types of drinkers")
```

Pie charts

We can also use the `pie()` function to create pie charts

```
> pie(drinks_table)
```



Questions?



Visualizing Quantitative Data

Visualizing quantitative data: histograms

A histogram is data visualization that allows us to which range of quantitative values occurs most frequently

The first few okcupid users' heights

```
> profiles$height[1:5]
```

75, 70, 68, 76, ...

To create a histogram we create a set of intervals

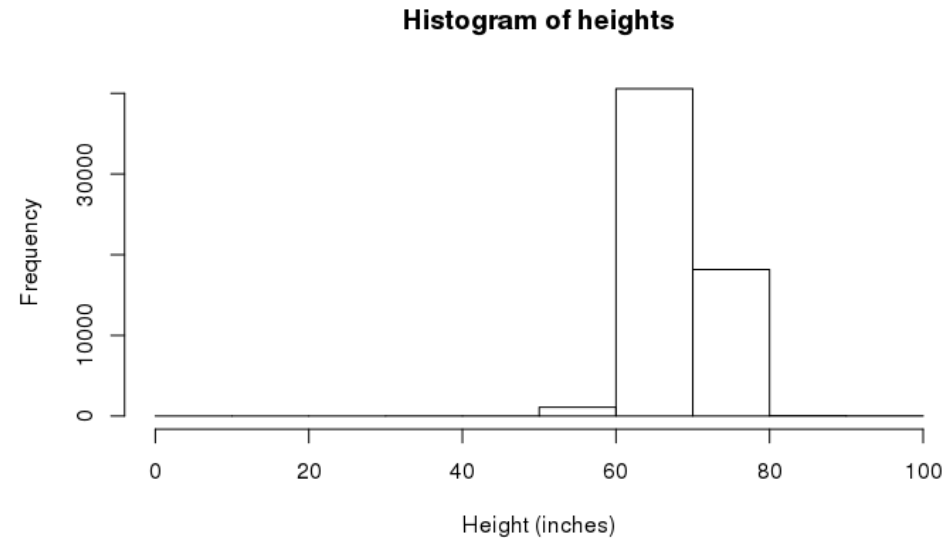
- (60-65], (65-70], (70-75], (75-80]

We count the number of points that fall in each interval

We create a bar chart with the counts in each bin

Histograms of heights

Height (inches)	Frequency Count
(0-10]	6
(10-20]	0
(20-30]	1
(30-40]	13
(40-50]	9
(50-60]	1097
(60-70]	40575
(70-80]	18164
(80-90]	50
>90	28



Visualizing heights

We can create histograms in R using the `hist()` function

Can you create a histogram of heights?

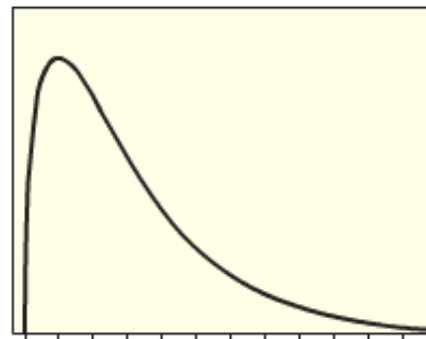
```
hist(profiles$height)
```

How can you add more bars to the histogram?

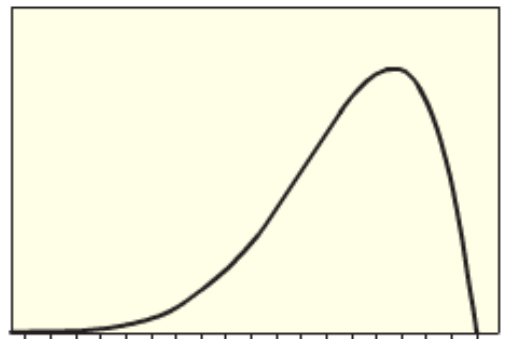
- How can we figure out how to add more bars to a histogram?

```
hist(profiles$height, breaks = 50)
```

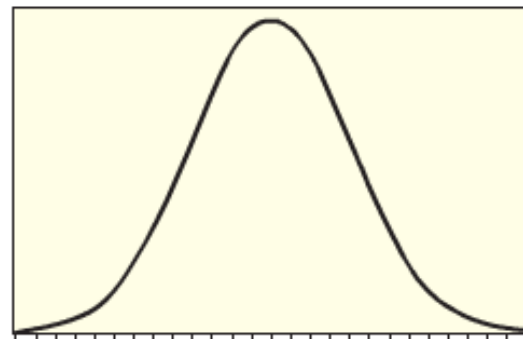
Common shapes for distributions



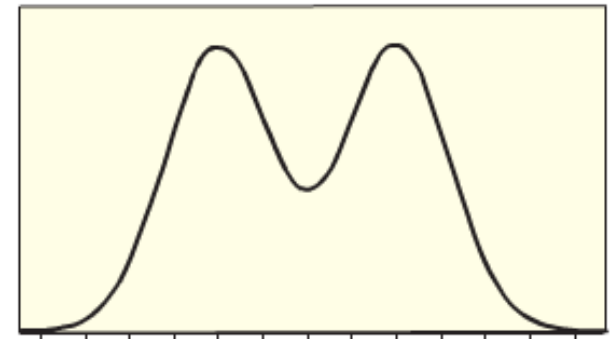
(a) Skewed to the right



(b) Skewed to the left



(c) Symmetric and bell-shaped



(d) Symmetric but not bell-shaped

Statistics for quantitative data

Measure of central tendency: The mean

$$\text{Mean} = \frac{\text{Sum of all data values}}{\text{Number of data values}}$$

$$\text{Mean} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

R: `mean(x)`

R: `mean(x, na.rm = TRUE)`

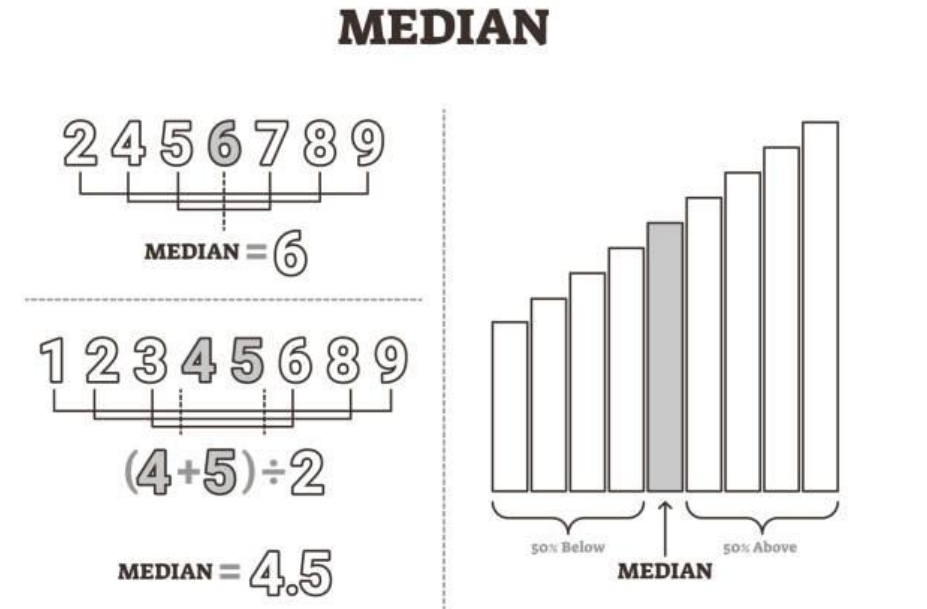
The median

The **median** is the value that splits the data in half

- i.e., half the values are less than the median and half are greater than the median

The median of a data set of size n is:

- If n is odd: The middle value of the sorted data
- If n is even: The average of the middle two values of the sorted data



R: `median(v)`
`median(v, na.rm = TRUE)`

Resistance

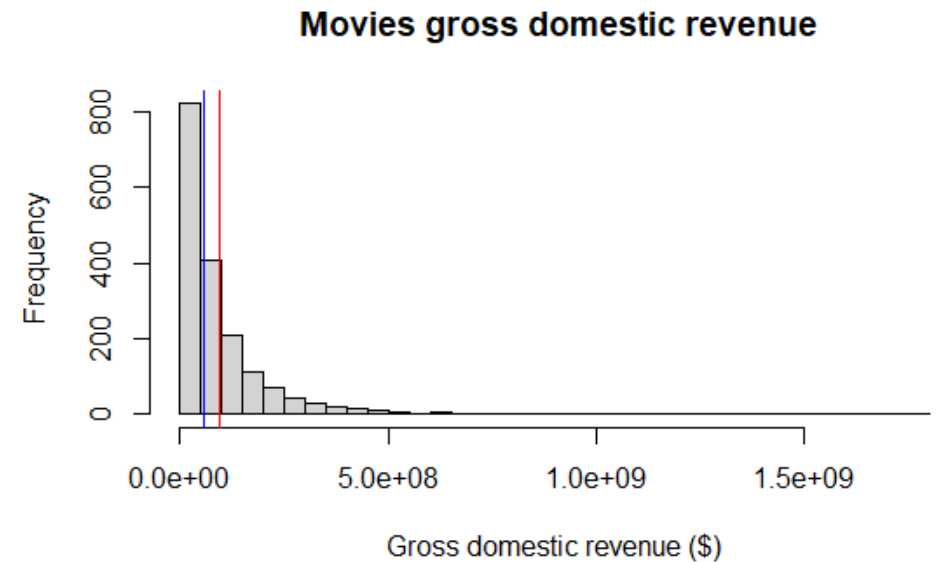
We say that a statistics is **resistant** if it is relatively unaffected by extreme values (outliers)

The median is resistant while the mean is not

Example:

Mean US salary = \$72,641

Median US salary = \$51,939



Measures of spread

Measure of spread 1: The standard deviation

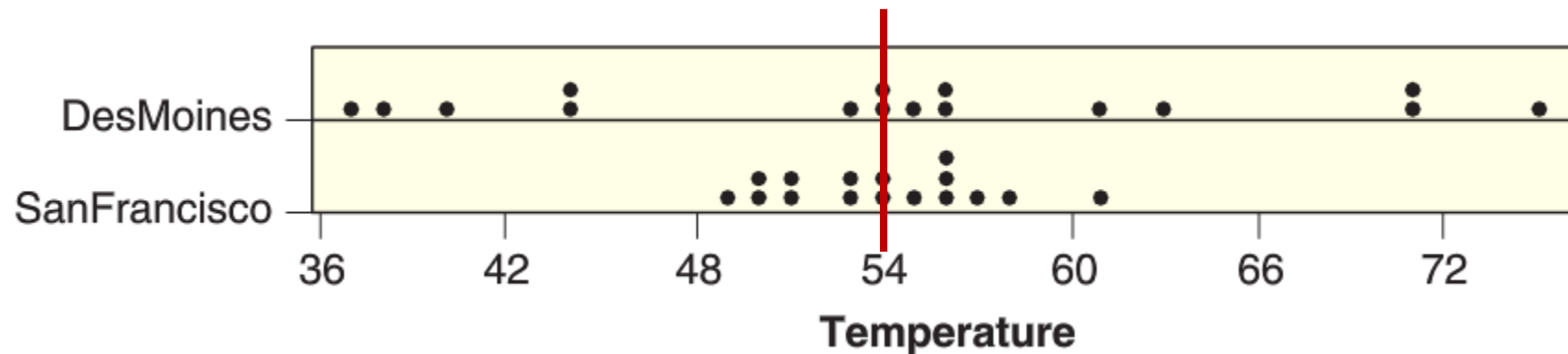
The **standard deviation** is a statistic that quantifies how far the data is spread

It can be computed using the following formula:

$$s = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Average monthly temperature: Des Moines vs. San Francisco

Data measured on April 14th from 1997 to 2010:



Mean temperature (°F): Des Moines = 54.49 San Francisco = 54.01
Standard deviation (°F): Des Moines = 11.73 San Francisco = 3.38

Example: computing the standard deviation

Suppose we had a sample with $n = 4$ points:

$$x_1 = 8, \quad x_2 = 2, \quad x_3 = 6, \quad x_4 = 4,$$

We can compute the mean using the formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{4} \cdot (x_1 + x_2 + x_3 + x_4) = \frac{1}{4} \cdot (8 + 2 + 6 + 4) = 5$$

The standard deviation can be computed using the formula:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (\text{remember order of operations!})$$

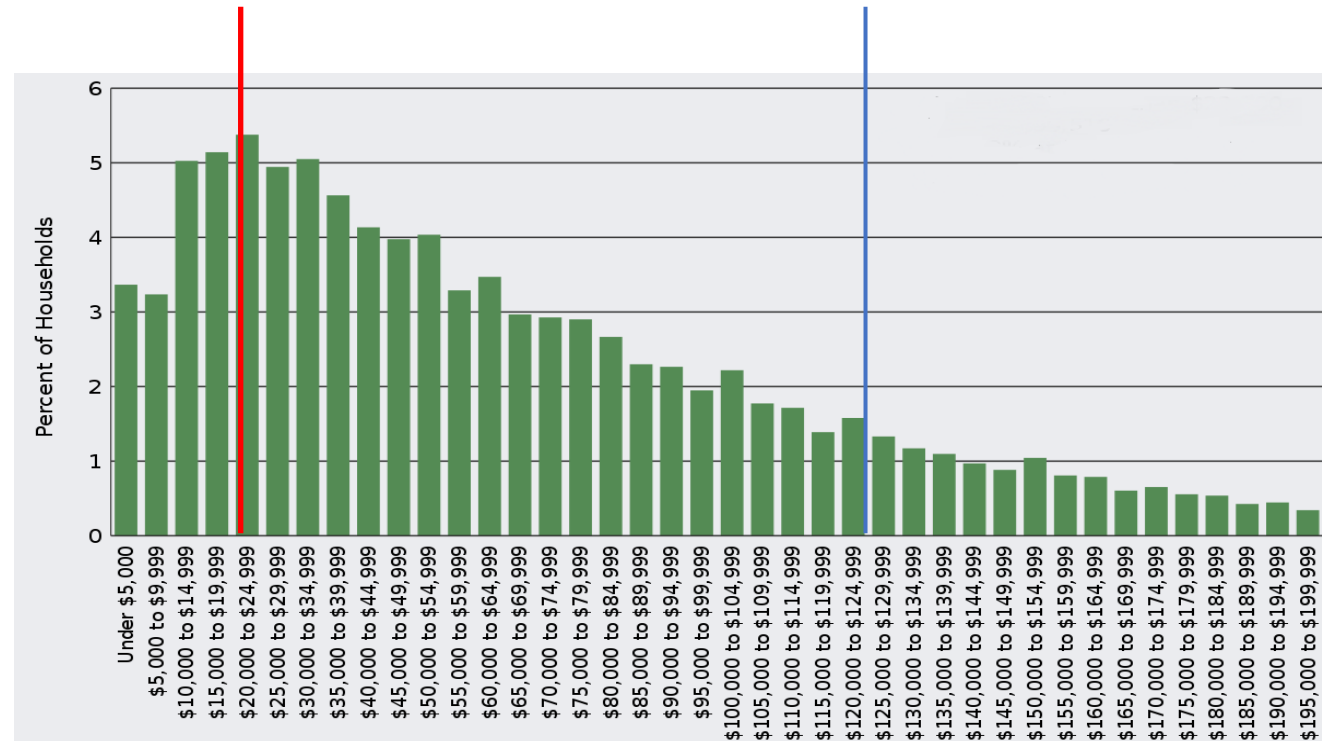
Percentiles

The **Pth percentile** is the value of a quantitative variable which is greater than P percent of the data

For the US income distribution what are the 20th and 80th percentiles?

20th percentile = \$21,430

80th percentile = \$112,254



R: `quantile(v, .95)`

Five Number Summary

A **five-number summary** is a set of five descriptive statistics that provides a concise overview of a dataset's distribution

Five Number Summary = (minimum, Q_1 , median, Q_3 , maximum)

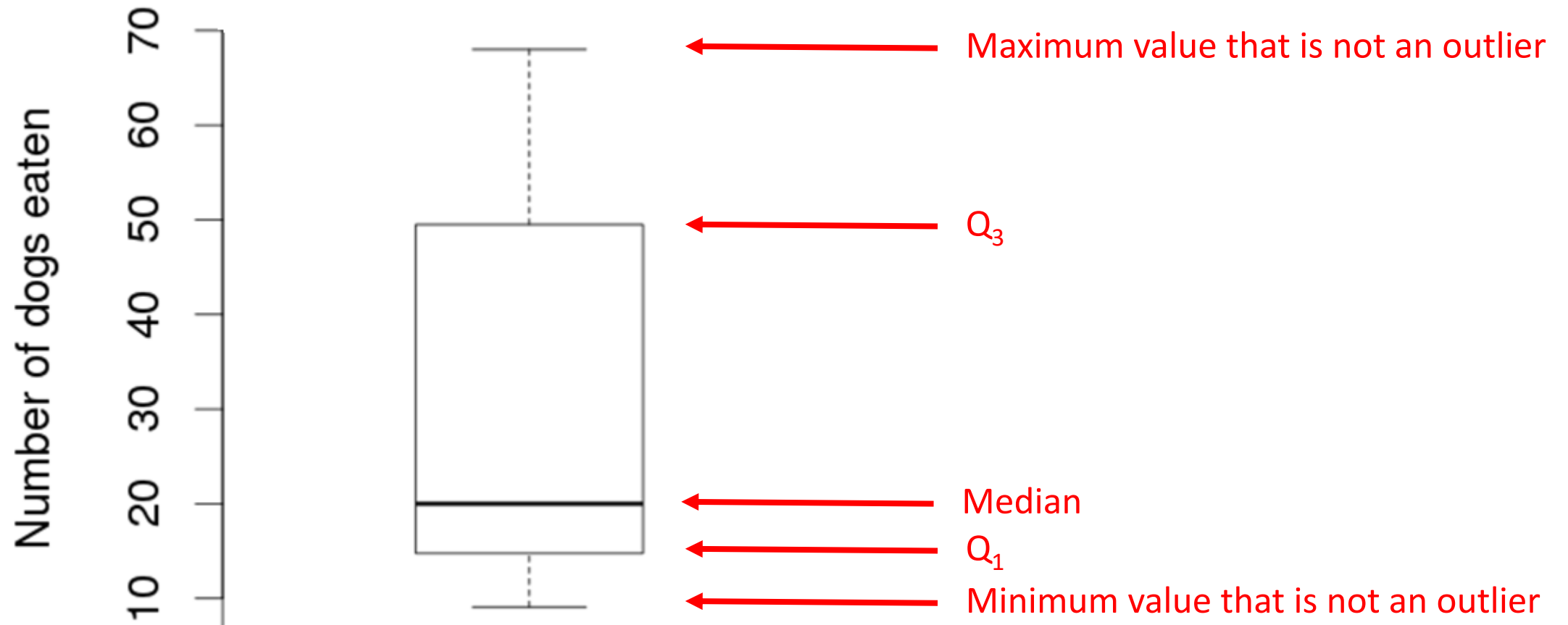
Q_1 = 25th percentile (also called 1st quartile)

Q_3 = 75th percentile (also called 3rd quartile)

Roughly divides the data into fourths

Measure of spread 2: Interquartile range (IQR) = $Q_3 - Q_1$

Box plots can also visualize quantitative data



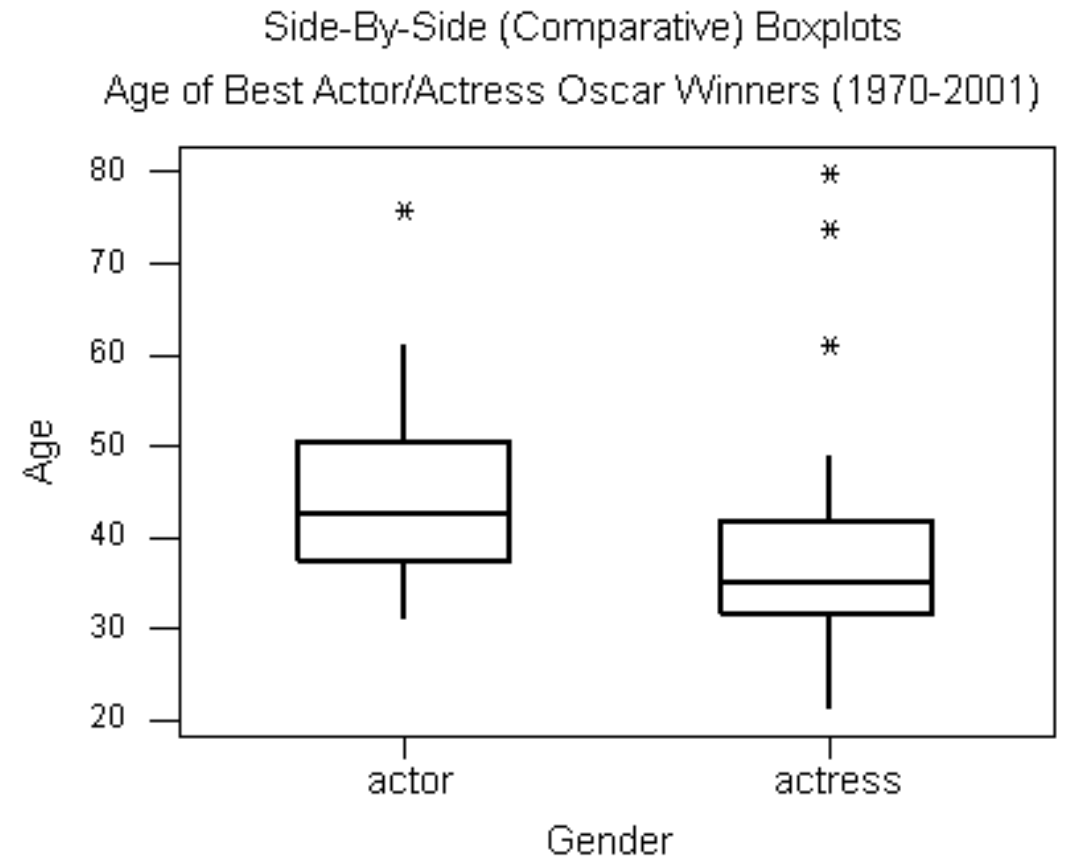
R: `boxplot(v)`

Side-by-side boxplots

Boxplots are particularly useful for comparing distributions!

Let's look at the ages that people won the best actor/actress Oscar

What does this figure tell us?



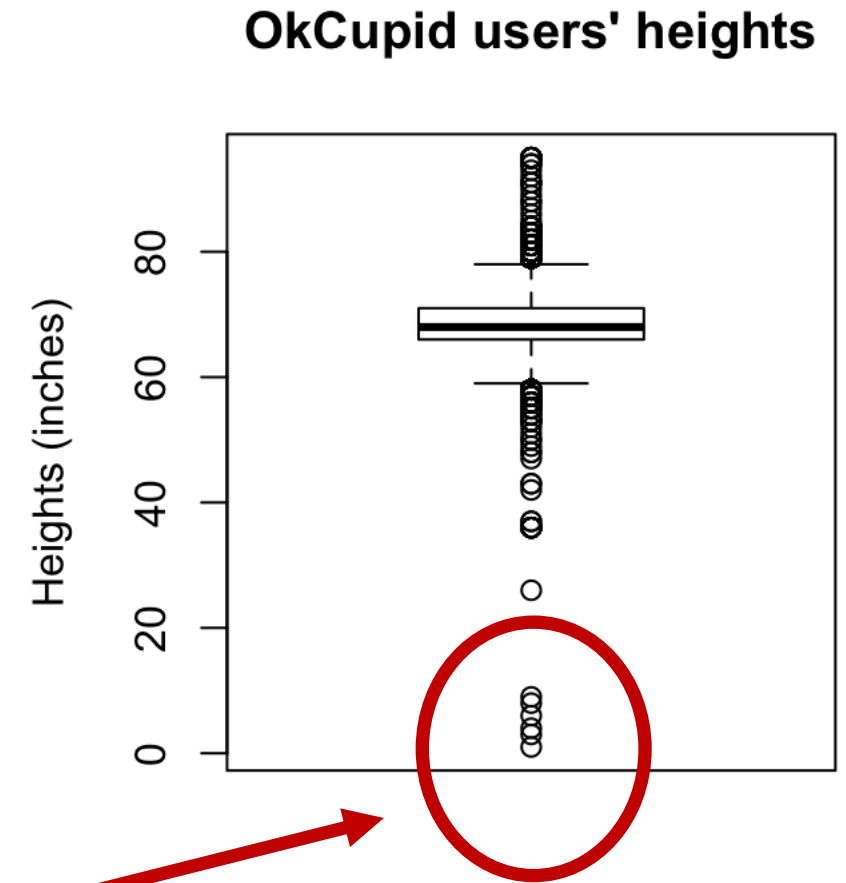
Outliers

Outliers on boxplots are values that are more than $1.5 * IQR$

What should we do if we have outliers?

Investigate!

- If there are due to an error, remove them



People under 20" tall?

Outliers

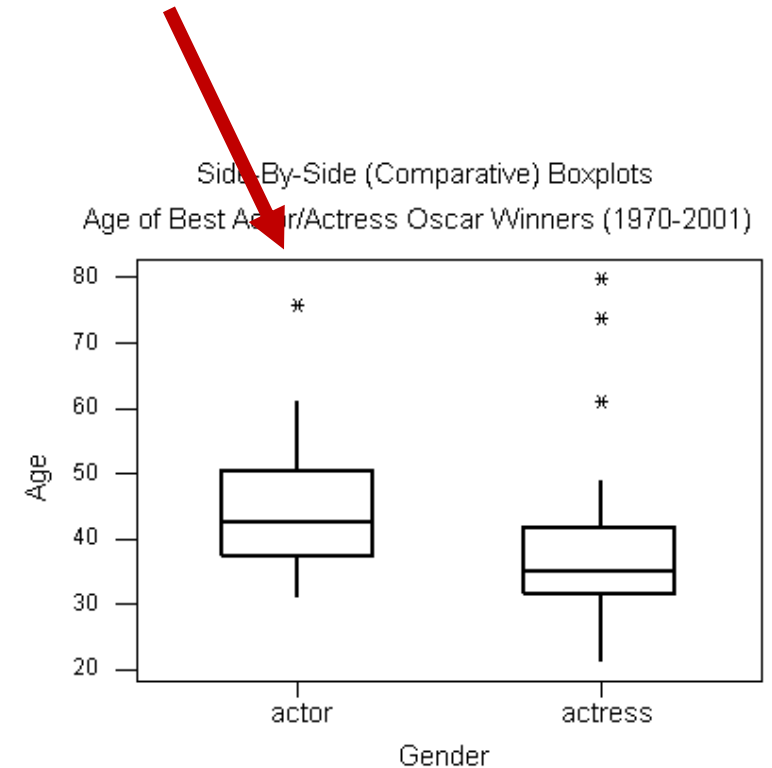
Outliers on boxplots are values that are more than $1.5 * IQR$

What should we do if we have outliers?

Investigate:

- If there are due to an error, remove them
- **If not, need to account for them**

Who is this actor?



Questions?



Let's try it in RStudio!

Visualizing two quantitative variables

CitiBike data

Let's look at the bike share data from NYC

```
> load('daily_bike_totals.rda')
```



[CitiBike analysis](#)

What does each case correspond to?

We can use the `dim()` function to get how many cases and variables there are

- How many are there?

Scatterplot

A **scatterplot** graphs the relationship between two variables

- Each axis represents the value of one variables
- Each point the plot shows the value for the two variables for a single data case

If there is an explanatory and response variable, then the explanatory variable is put on the x-axis and the response variable is put on the y-axis.

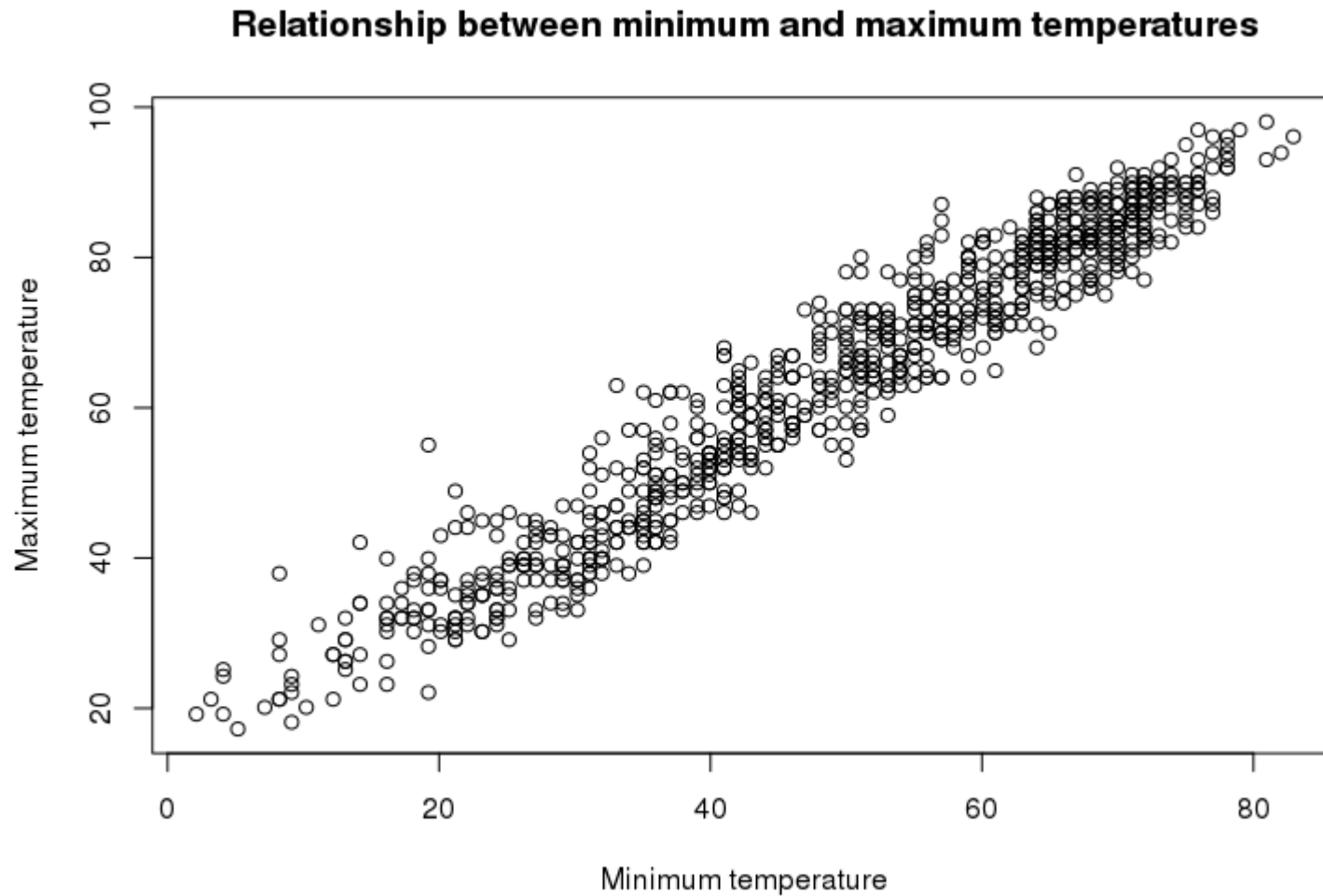
Scatter plots

We can use the `plot(x, y)` function to create scatter plots

Can you create a scatter plot of the relationship between the minimum and maximum temperatures?

```
plot(bike_daily_data$min_temperature,  
     bike_daily_data$max_temperature,  
     xlab = "Minimum temperature",  
     ylab = "Maximum temperature",  
     main = "Relationship between min and temp")
```

Scatter plots



Plotting time series

We can use the `plot(x, y)` function to plot time series

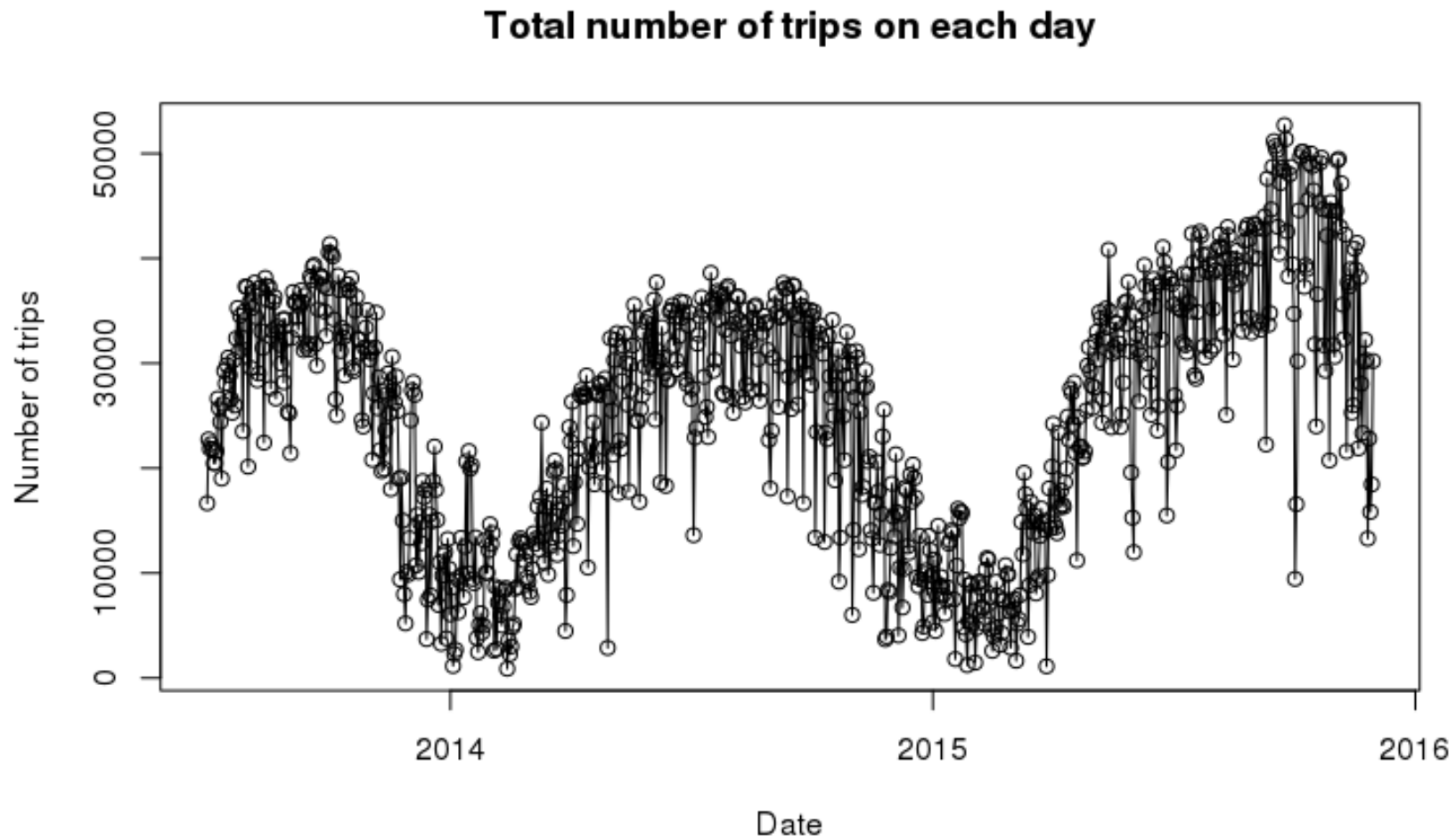
we can connect the points in a plot using

```
plot(x, y, type = 'l') # line graph
```

```
plot(x, y, type = 'o') # both points and a line
```

```
plot(bike_daily_data$date, bike_daily_data$trips,  
     type = 'o',  
     xlab = "Date",  
     ylab = "Number of trips",  
     main = "Total number of trips on each day")
```

Plotting time series



The correlation coefficient

The **correlation** is measure of the strength and direction of a linear association between two variables

$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

R: `cor(x, y)`

Properties of the correlation

Correlation is always between -1 and 1: $-1 \leq r \leq 1$

The sign of r indicates the direction of the association

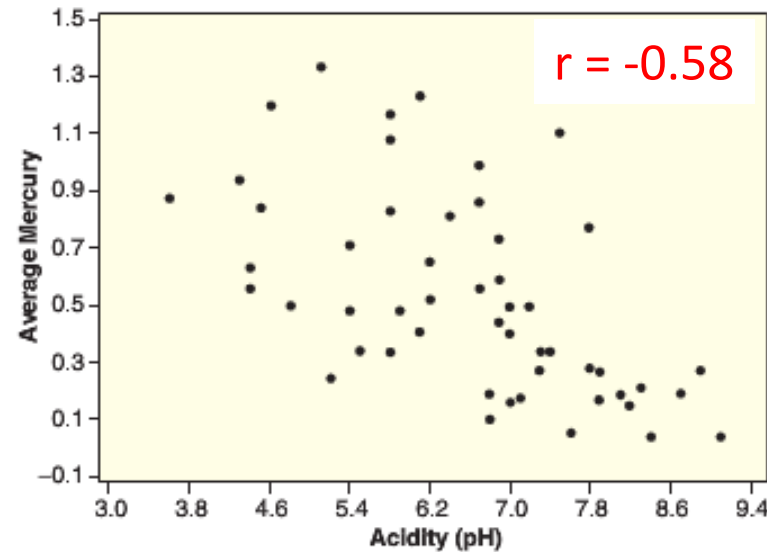
Values close to ± 1 show strong linear relationships, values close to 0 show no linear relationship

Correlation is symmetric: $r = \text{cor}(x, y) = \text{cor}(y, x)$

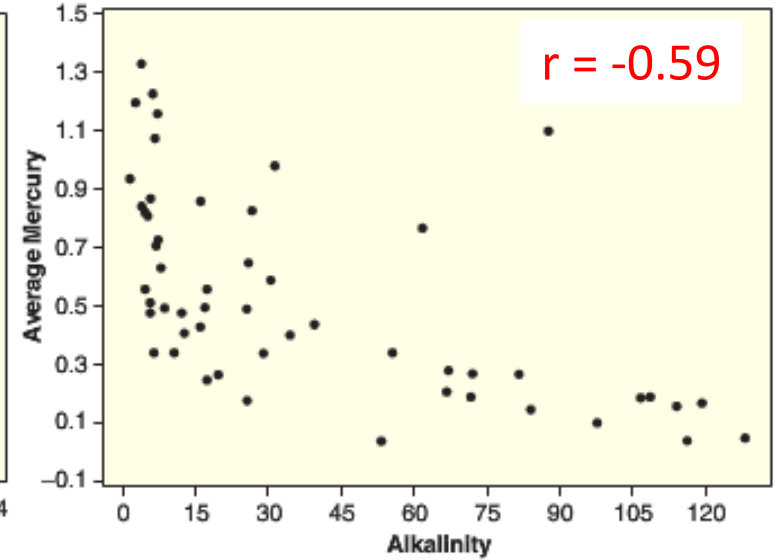
$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Florida lakes

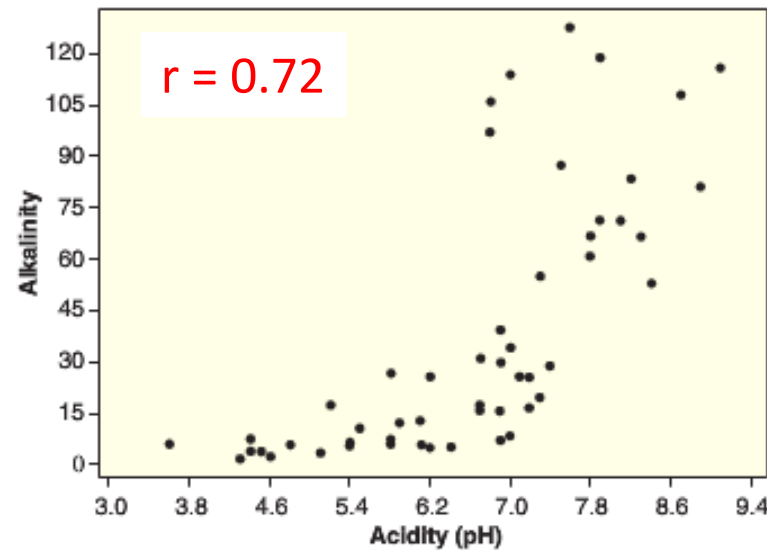
Correlation game



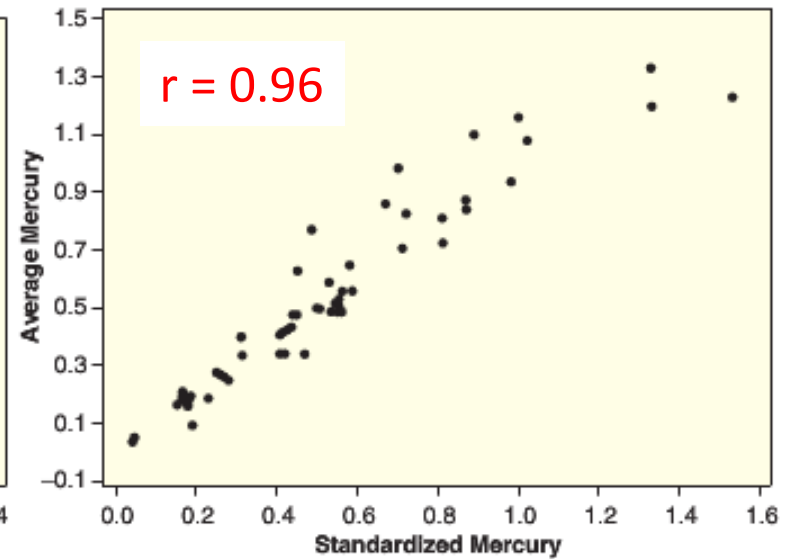
(a) Average mercury level vs acidity



(b) Average mercury level vs alkalinity



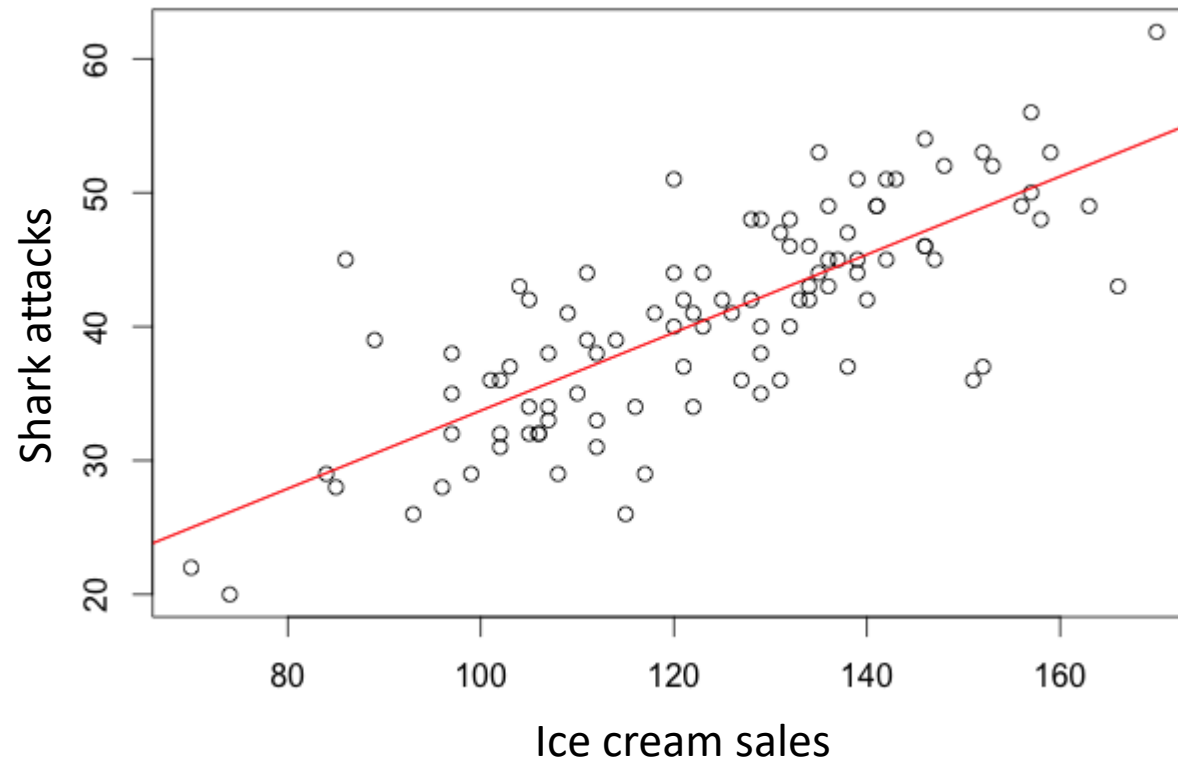
(c) Alkalinity vs acidity



(d) Average vs standardized mercury levels

Correlation caution #1

A strong positive or negative correlation does not (necessarily) imply a cause and effect relationship between two variables



Next class: data transformations with dplyr